

Week 4

Essential Python Tools for Data Science

Equipping students with core Python tools and libraries required for large-scale data analysis in AI.

Learning Objectives

- Understand the basics of Python for data science workflows
- Identify and use essential libraries (pandas, numpy, matplotlib, seaborn)
- Practice working with Jupyter Notebooks for interactive coding and analysis
- Prepare for hands-on exploratory data analysis in future weeks

Session Agenda

1. Why Python; The industry standard for data science and AI
2. Key libraries overview; pandas, numpy, matplotlib, seaborn
3. Setting up the Jupyter environment
4. Practical demonstration; Importing and using libraries
5. Activity; Install tools and run your first Notebook
6. Industry context; Real-world applications and expectations
7. Review and assessment preparation

Why Python for Data Science

- Python is widely used in AI, machine learning, and data analysis roles
- Large open-source community and extensive library support
- Readable, flexible syntax makes it ideal for rapid prototyping and large-scale data tasks
- Skills with Python are valuable in AI research labs and industry projects

Introduction to Jupyter Notebooks

- Jupyter Notebooks provide an interactive coding environment for data exploration and visualization
- Combine code, visualizations, and documentation in a single, shareable document
- Widely used in research, industry, and collaborative projects
- Enables reproducible workflows and documentation best practices

Essential Library 1; pandas

- Purpose; Data manipulation and analysis with structured data (tables, csv, excel, databases)
- Key features; DataFrame object, data filtering, grouping, cleaning, and aggregation
- Example;

```
import pandas as pd  
df = pd.read_csv('data.csv')  
df.head()
```

Essential Library 2; numpy

- Purpose; Fast, efficient numerical computing with large datasets
- Key features; Multidimensional arrays, mathematical operations, support for linear algebra
- Example;

```
import numpy as np
array = np.array([1, 2, 3, 4])
print(array.mean())
```

Essential Libraries 3 & 4; matplotlib and seaborn

- matplotlib; Foundation library for plotting and basic data visualization
- seaborn; Builds on matplotlib to provide advanced, aesthetically pleasing statistical plots
- Use cases; Plotting distributions, trends, and relationships in data
- Example;

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df['column_name'])
plt.show()
```


Practical Activity; Setup and First Steps

- Install Anaconda or Miniconda (recommended) or Python 3 with pip
- Install required libraries; pandas, numpy, matplotlib, seaborn, jupyter
- Launch Jupyter Notebook; Test by importing libraries and displaying their versions
- Example;

```
import pandas, numpy, matplotlib, seaborn  
print(pandas.__version__)
```

Industry Context; Why These Tools Matter

- Tools covered today are the backbone of most AI and large-scale data projects
- Used by companies such as Google, Meta, and open-source consortia working with LLMs and generative models
- Core skills for any data science or AI engineering career
- Enable efficient handling and analysis of real-world datasets such as LAION-5B, COCO, or UCI repositories

Link to Previous Weeks

- Week 1; The importance of large open datasets in AI
- Week 2; Where to find reliable, open datasets for analysis
- Week 3; Foundations of data ethics and legal responsibilities in handling data
- This week builds foundation for using and analyzing open datasets introduced earlier

Assessment Preparation

- Mastery of pandas, numpy, and Jupyter is essential for:
 - Exploratory Data Analysis assessment (due Week 14)
 - Machine Learning project (due Week 18)
- Practicing data importing, cleaning, and basic visualization will prepare you for analyzing real-world AI datasets
- Review Week 4 code examples and repeat activities using a dataset of your choice

Next Steps

- For Week 5: You will use these Python tools to explore and document open datasets
- Prepare Jupyter Notebooks with exploratory code ready to share
- Review documentation for pandas and seaborn to deepen your understanding

Questions or challenges; bring them to class or raise them with your lecturer

